



Pranav Khare <khare.pranav@gmail.com>

Invitation to Review for IEEE Access Access-2024-51445

1 message

IEEE Access <onbehalf@manuscriptcentral.com>

Sun, Feb 2, 2025 at 9:25 PM

Reply-To: ieeeaccess2@ieee.org

To: khare.pranav@gmail.com

03-Feb-2025

Dear Mr. Khare:

The manuscript entitled "Deterministic Compression of Word Embeddings" has been submitted to IEEE Access. Since your expertise is relevant to this article, we would be grateful if you would consider reviewing it.

The abstract appears at the end of this letter, along with the names of the authors. Please let me know within 2 days if you will be able to accept my invitation to review. If you agree to review, we ask that you provide your review comments within **7 days** to assist IEEE Access in maintaining its expedited peer review process of 4 to 6 weeks. If you are interested in reviewing but need additional time to complete the review, we can grant a reasonable deadline extension upon request. For additional information on IEEE Access and tips on completing a quality review, please visit our [Reviewer Best Practices guide](#).

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Once you accept my invitation to review this manuscript, you will be notified via email with directions on how to access the manuscript and the reviewer instructions in the Reviewer Center.

Thank you for your assistance with the peer review of this IEEE Access article.

Sincerely,

Prof. Mehrdad Saif
Editor-in-Chief, IEEE Access
ieeeaccess2@ieee.org

MANUSCRIPT DETAILS

TYPE: Methods

TITLE: Deterministic Compression of Word Embeddings

AUTHORS: Nakamura, Yuki; SUZUKI, JUN; ITO, TAKUMI; INUI, KENTARO

ABSTRACT: Word embeddings are an indispensable technology in the field of artificial intelligence, particularly when text serves as the input for neural models. To further enhance their usability, several studies have tackled the compression of word embeddings while maintaining task performance. In the same vein, this study proposes a word embedding compression method that vastly differs from recent neural compression methods. A notable characteristic of the proposed method is that it guarantees a stable and reproducible compressed representation under identical configurations. The central idea of the proposed method is utilizing a deterministic procedure and a convex optimization. More specifically, we leverage the property of obtaining identical solutions for identical convex optimization problems to construct a deterministic procedure throughout the entire compression process. We conduct experiments on both intrinsic and extrinsic evaluation with various word embeddings. Our experimental results show the effectiveness of the proposed method in terms of the performance vs. compression ratio trade-off and the ability to offer reproducible results. Notably, the results of the machine translation experiment on the extrinsic evaluation show that the proposed method compresses the word embedding layer of the LSTM Decoder by a factor of 258 with a 0.2 point decrease in BLEU Score.

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